

Azure Hybrid Cloud - Hub & Spoke with On-Premises VPN Connectivity

Complete End-to-End Documentation

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# 🌐 Azure Hybrid Cloud Architecture
## Hub-and-Spoke with On-Premises VPN Connectivity
### Complete Implementation Guide

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## 1. Architecture Overview

### 1.1 What We Are Building

This documentation covers the complete implementation of a
Hybrid Cloud Architecture that connects:

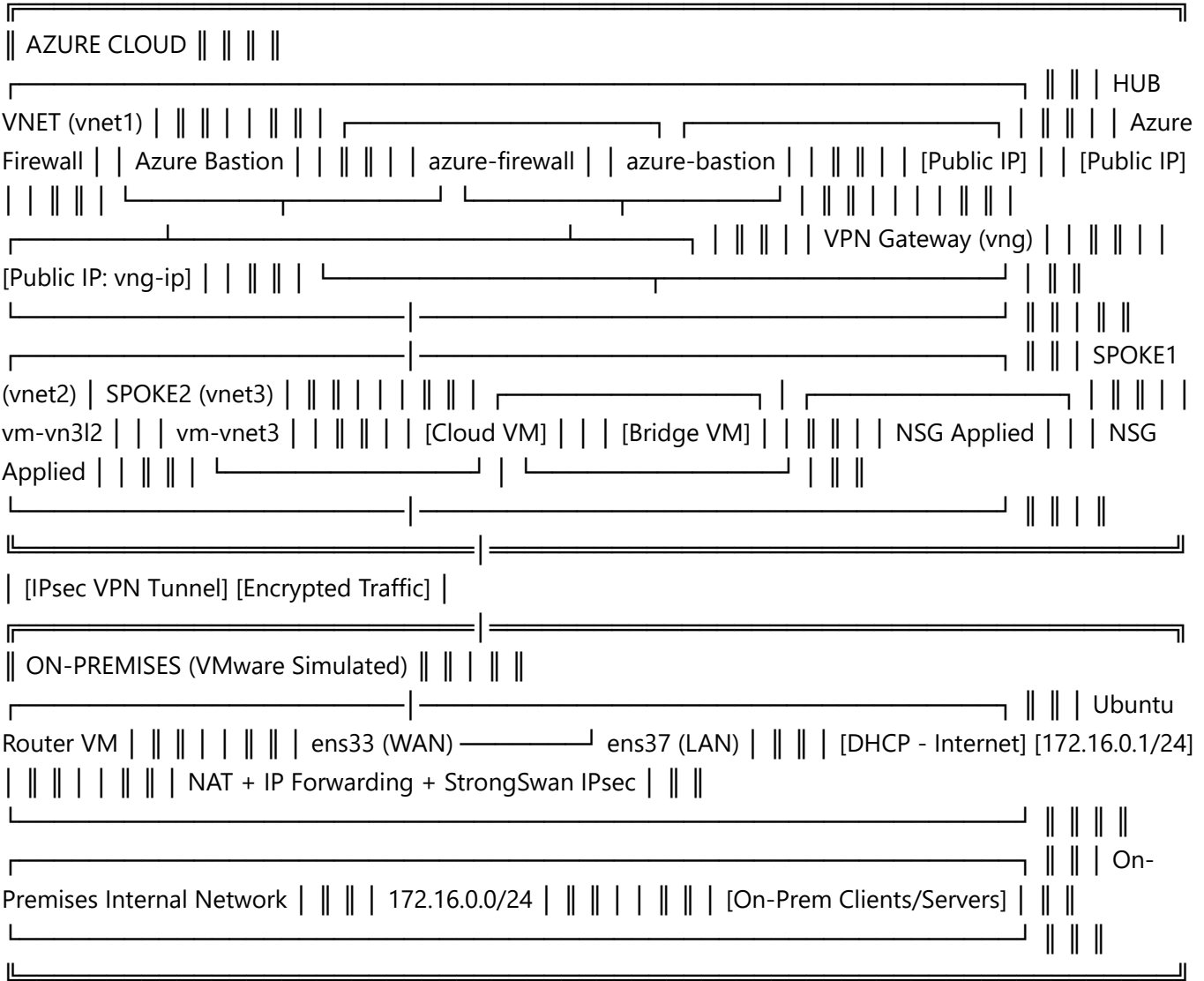


- ✓ Azure Hub-and-Spoke Network
- ✓ On-Premises Simulated Network (VMware Ubuntu)
- ✓ Secure Site-to-Site VPN Tunnel
- ✓ Centralized Security via Azure Firewall
- ✓ Zero-Trust Access via Azure Bastion



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1.2 High-Level Architecture Diagram



1.3 Traffic Flow Diagrams
Cloud-to-Cloud Communication

vm-vn3l2 → vnet2 → route2 → azure-firewall → vnet1/vnet3

Cloud-to-On-Premises Communication

vm-vn3l2 → vnet2 → azure-firewall → vng → VPN Tunnel → Ubuntu Router → 172.16.0.0/24

```
#### Administrative Access
```

Administrator → Internet → azure-bastion → vm-vn3l2 / vm-vnet3

```
#### Outbound Internet Traffic
```

vm-vn3l2 → azure-firewall → azure-firewall-ip → Internet

```
#### On-Premises to Azure Communication
```

On-Prem Client → Ubuntu Router (ens37) → StrongSwan → ens33 → VPN Tunnel → vng →
azure-firewall → Azure VMs

1.4 Component Summary Table

Component	Type	Purpose	Location
-----	-----	-----	-----
`vnet1`	Azure VNet	Hub Network	Azure
`vnet2`	Azure VNet	Spoke - Cloud Workloads	Azure
`vnet3`	Azure VNet	Spoke - Bridge to On-Prem	Azure
`azure-firewall`	Azure Firewall	Central Traffic Inspection	Hub (vnet1)
`azure-bastion`	Azure Bastion	Secure Remote Access	Hub (vnet1)
`vng`	VPN Gateway	Site-to-Site VPN	Hub (vnet1)
`vm-vn3l2`	Azure VM	Cloud Workload VM	vnet2
`vm-vnet3`	Azure VM	Bridge/DMZ VM	vnet3
`vm-vn3l2-nsg`	NSG	Cloud VM Security	vnet2
`vm-vnet3-nsg`	NSG	Bridge VM Security	vnet3
`route2`	Route Table	Traffic Routing	vnet2
`route3`	Route Table	On-Prem Traffic Routing	vnet3
Ubuntu Router	VMware VM	On-Prem Router/VPN	On-Premises

2. Prerequisites

2.1 Azure Requirements

✓ Azure Subscription (with contributor access) ✓ Resource Group created ✓ Azure Region selected (consistent across all resources) ✓ Sufficient vCPU quota for VMs ✓ VPN Gateway SKU support (VpnGw1 minimum)

2.2 On-Premises Requirements

✓ VMware Workstation / VMware Player installed ✓ Ubuntu Server 22.04 LTS ISO downloaded ✓

Minimum VM specs:

- CPU : 2 vCPUs
- RAM : 2 GB
- Disk : 20 GB
- NICs : 2 (NAT + Host-Only or NAT + NAT) ✓ Internet connectivity on host machine

2.3 Network Planning

Azure VNets: | vnet1 (Hub) : 10.1.0.0/16
| | vnet2 (Spoke1) : 10.2.0.0/16 | | vnet3 (Spoke2) : 10.3.0.0/16 | | | GatewaySubnet : 10.1.0.0/27 | |
AzureFirewall : 10.1.1.0/26 | | AzureBastion : 10.1.2.0/27 |
|

On-Premises: | LAN Network :
172.16.0.0/24 | | Router LAN IP : 172.16.0.1 | | Router WAN IP : DHCP (from VMware) |
|

3. Azure Infrastructure Setup

3.1 Resource Group

Azure Portal → Resource Groups → Create

Name : rg-hybrid-network Region : East US (or your preferred region) Tags : Environment=Production, Project=HybridCloud

```
### 3.2 Virtual Networks

#### Hub VNet (vnet1)
```

Azure Portal → Virtual Networks → Create

Basic: Name : vnet1 Region : East US Resource Group : rg-hybrid-network

IP Addresses: Address Space : 10.1.0.0/16

Subnets:		Subnet Name	Address
Range		GatewaySubnet	
10.1.0.0/27	AzureFirewallSubnet	10.1.1.0/26	AzureBastionSubnet
10.1.2.0/27	hub-general-subnet	10.1.3.0/24	

```
#### Spoke VNet (vnet2)
```

Azure Portal → Virtual Networks → Create

Basic: Name : vnet2 Region : East US Resource Group : rg-hybrid-network

IP Addresses: Address Space : 10.2.0.0/16

Subnets:		Subnet Name	Address
Range		workload-subnet	
10.2.1.0/24			

```
#### Spoke VNet (vnet3)
```

Azure Portal → Virtual Networks → Create

Basic: Name : vnet3 Region : East US Resource Group : rg-hybrid-network

IP Addresses: Address Space : 10.3.0.0/16

Subnets:		Subnet Name	Address
Range		onprem-bridge-subnet	
10.3.1.0/24			

```
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```

3.3 VNet Peering (Hub-Spoke)

Peering 1: vnet1 ↔ vnet2 ————— Azure
Portal → vnet1 → Peerings → Add

Peering Name (vnet1→vnet2) : hub-to-spoke1 Remote VNet : vnet2 Peering Name (vnet2→vnet1) : spoke1-to-hub

Settings: ☒ Allow gateway transit (on vnet1 side) ☒ Use remote gateway (on vnet2 side) ☒ Allow forwarded traffic

Peering 2: vnet1 ↔ vnet3 ————— Peering
Name (vnet1→vnet3) : hub-to-spoke2 Remote VNet : vnet3 Peering Name (vnet3→vnet1) : spoke2-to-hub

Settings: ☒ Allow gateway transit (on vnet1 side) ☒ Use remote gateway (on vnet3 side) ☒ Allow forwarded traffic

3.4 Azure Firewall

Azure Portal → Firewalls → Create

Basic: Name : azure-firewall Region : East US Resource Group : rg-hybrid-network SKU : Standard VNet : vnet1 Subnet : AzureFirewallSubnet (10.1.1.0/26) Public IP : Create New → azure-firewall-ip

Firewall Policy Rules

Azure Portal → Firewall Policies → Create → azure-fw-policy

Network Rules:

Rule Collection: allow-spoke-to-spoke Priority: 100									
Name	Protocol	Source	Dest	Dest Port					
allow-vnet2	Any	10.2.0.0	10.3.0.0	*	to-vnet3	/16	/16		
allow-vnet2	Any	10.2.0.0	172.16.0	*	to-onprem	/16	.0/24		

allow-onprem		Any		172.16.0		10.0.0.0		*			to-azure			.0/24		/8		

Application Rules:

Rule Collection: allow-internet Priority: 200														
Name	Protocol	Source	Target FQDN											
allow-web	HTTP/S	10.0.0.0	*				/8							

```
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### 3.5 Azure Bastion
```

Azure Portal → Bastions → Create

Basic: Name : azure-bastion Region : East US Resource Group : rg-hybrid-network VNet : vnet1 Subnet : AzureBastionSubnet (10.1.2.0/27) Public IP : Create New → azure-bastion-ip SKU : Basic

```
---

### 3.6 VPN Gateway
```

Azure Portal → Virtual Network Gateways → Create

Basic: Name : vng Region : East US Gateway Type : VPN VPN Type : Route-based SKU : VpnGw1 Generation : Generation1 VNet : vnet1 Subnet : GatewaySubnet (10.1.0.0/27) Public IP : Create New → vng-ip Active-Active : Disabled BGP : Disabled

⚠ NOTE: VPN Gateway takes 30-45 minutes to deploy

```
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### 3.7 Route Tables

#### Route Table for vnet2 (route2)
```

Azure Portal → Route Tables → Create

Name : route2 Region : East US Resource Group : rg-hybrid-network

Routes:

Route Name	Address Prefix	Next Hop Type	Next Hop						
to-hub-firewall	0.0.0.0/0	VirtualAppl.	10.1.1.4		to-onprem	172.16.0.0/24	VirtualAppl.	10.1.1.4	
to-vnet3	10.3.0.0/16	VirtualAppl.	10.1.1.4						

Associate to: vnet2 → workload-subnet

```
#### Route Table for vnet3 (route3)
```

Azure Portal → Route Tables → Create

Name : route3 Region : East US Resource Group : rg-hybrid-network

Routes:

Route Name	Address Prefix	Next Hop Type	Next Hop						
to-hub-firewall	0.0.0.0/0	VirtualAppl.	10.1.1.4		to-onprem	172.16.0.0/24	VirtualAppl.	10.1.1.4	to-vnet2
	10.2.0.0/16	VirtualAppl.	10.1.1.4						

Associate to: vnet3 → onprem-bridge-subnet

```
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### 3.8 Network Security Groups

#### NSG for Cloud VM (vm-vn312-nsg)
```

Azure Portal → Network Security Groups → Create

Name: vm-vn312-nsg

Inbound Rules:

Priority	Name	Port	Protocol	Source	Action						
100	allow-bastion	3389	TCP	Bastion	Allow					22	TCP Subnet

200	allow-internal	*	Any	10.0.0	Allow					.0/8	
4096	deny-all	*	Any	*	Deny						

Associate to: vm-vn3l2 network interface

```
##### NSG for Bridge VM (vm-vnet3-nsg)
```

Name: vm-vnet3-nsg

Inbound Rules:

Priority	Name	Port	Protocol	Source	Action						
100	allow-bastion	22	TCP	Bastion	Allow					Subnet	
200	allow-vpn	*	Any	172.16	Allow					.0/24	
300	allow-internal	*	Any	10.0.0	Allow					.0/8	
4096	deny-all	*	Any	*	Deny						

Associate to: vm-vnet3 network interface

```
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### 3.9 Azure Virtual Machines

#### Cloud VM (vm-vn3l2)
```

Azure Portal → Virtual Machines → Create

Basic: Name : vm-vn3l2 Region : East US Image : Ubuntu Server 22.04 LTS Size : Standard_B2s Auth Type : SSH Public Key Username : azureuser

Networking: VNet : vnet2 Subnet : workload-subnet Public IP : None (access via Bastion) NSG : vm-vn3l2-nsg

Disks: OS Disk : vm-vn3l2_disk1_a75e8e95... Type : Standard SSD

```
#### Bridge VM (vm-vnet3)
```

Azure Portal → Virtual Machines → Create

Basic: Name : vm-vnet3 Region : East US Image : Ubuntu Server 22.04 LTS Size : Standard_B2s Auth Type : SSH
Public Key Username : azureuser

Networking: VNet : vnet3 Subnet : onprem-bridge-subnet Public IP : None (access via Bastion) NSG : vm-vnet3-nsg

Disks: OS Disk : vm-vnet3_disk1_3bea63b1... Type : Standard SSD

```
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## 4. On-Premises Router VM Setup (VMware)

### 4.1 VMware VM Configuration
```

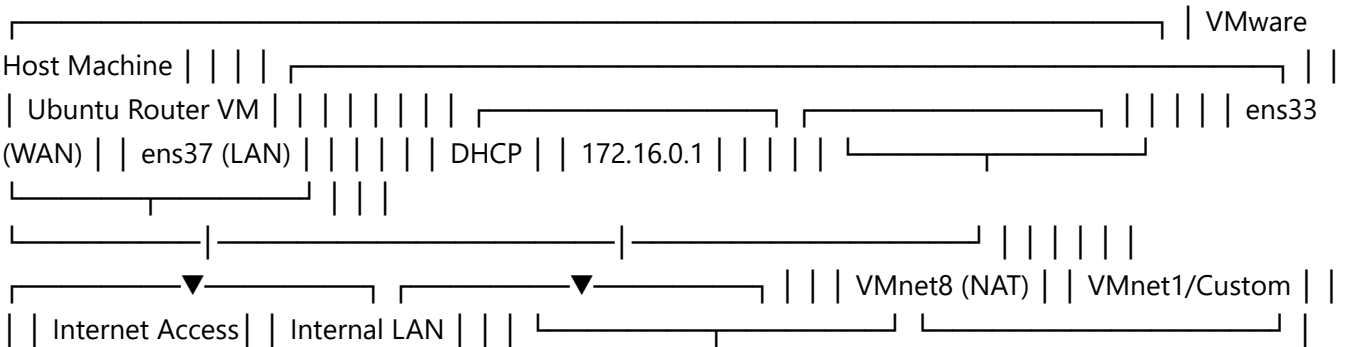
VMware Workstation → Create New Virtual Machine

VM Settings:

OS : Ubuntu Server 22.04 LTS | CPU : 2 vCPUs | RAM : 2048 MB | Disk : 20 GB | Network Adapters: | NIC 1 (ens33) : NAT | NIC 2 (ens37) : NAT / Host-Only |

⚠ IMPORTANT: Two NICs are required NIC1 = WAN (Internet/Azure connectivity) NIC2 = LAN (On-premises internal network)

```
### 4.2 VMware Network Adapter Layout
```




```
# /etc/netplan/00-installer-config.yaml
# -----
# On-Premises Router VM - Network Configuration
#
# ens33 → WAN Interface (NAT → Internet → Azure VPN)
# ens37 → LAN Interface (On-premises internal network)
# -----

network:
  version: 2
  renderer: networkd

ethernets:

  # — WAN Interface -----
  ens33:
    dhcp4: true
    dhcp4-overrides:
      route-metric: 100      # Preferred default route
    nameservers:
      addresses:
        - 8.8.8.8
        - 8.8.4.4

  # — LAN Interface -----
  ens37:
    dhcp4: false
    addresses:
      - 172.16.0.1/24      # Router LAN IP
    nameservers:
      addresses:
        - 8.8.8.8
        - 8.8.4.4
```

```
# Step 4: Validate configuration (no output = no errors)
sudo netplan generate

# Step 5: Apply configuration
sudo netplan apply

# Step 6: Verify routing table (should show 3 routes)
ip route show

# Expected Output:
# -----
# default via 192.168.x.x dev ens33 proto dhcp metric 100
# 172.16.0.0/24 dev ens37 proto kernel scope link src 172.16.0.1
# 192.168.x.0/24 dev ens33 proto kernel scope link src 192.168.x.x
# -----
```

5.4 Enable IP Forwarding

```
# Step 7: Create sysctl configuration
sudo nano /etc/sysctl.d/99-onprem-router.conf
```

```
# /etc/sysctl.d/99-onprem-router.conf
# _____
# IP Forwarding & Routing Configuration
# _____

# Enable IPv4 packet forwarding (required for routing)
net.ipv4.ip_forward = 1

# Disable ICMP redirects (security best practice for routers)
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0

# Disable reverse path filtering (required for VPN)
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.all.rp_filter = 0
```

```
# Step 8: Apply sysctl settings
sudo sysctl --system

# Step 9: Verify IP forwarding is enabled
sysctl net.ipv4.ip_forward
# Expected: net.ipv4.ip_forward = 1
```

5.5 Configure NAT with iptables

```
# Step 10: Add NAT masquerade (internet access for LAN clients)
sudo iptables -t nat -A POSTROUTING -o ens33 -j MASQUERADE

# Step 11: Allow forwarding between interfaces
sudo iptables -A FORWARD -i ens37 -o ens33 -j ACCEPT
sudo iptables -A FORWARD -i ens33 -o ens37 \
    -m state --state RELATED,ESTABLISHED -j ACCEPT

# Step 12: Verify NAT rules
sudo iptables -t nat -L -n -v

# Step 13: Save rules permanently
```

```
sudo apt install iptables-persistent -y
sudo netfilter-persistent save
```

5.6 Verify Internet Connectivity

```
# Step 14: Test internet connectivity
ping -c 4 8.8.8.8

# Step 15: Test DNS resolution
ping -c 4 google.com

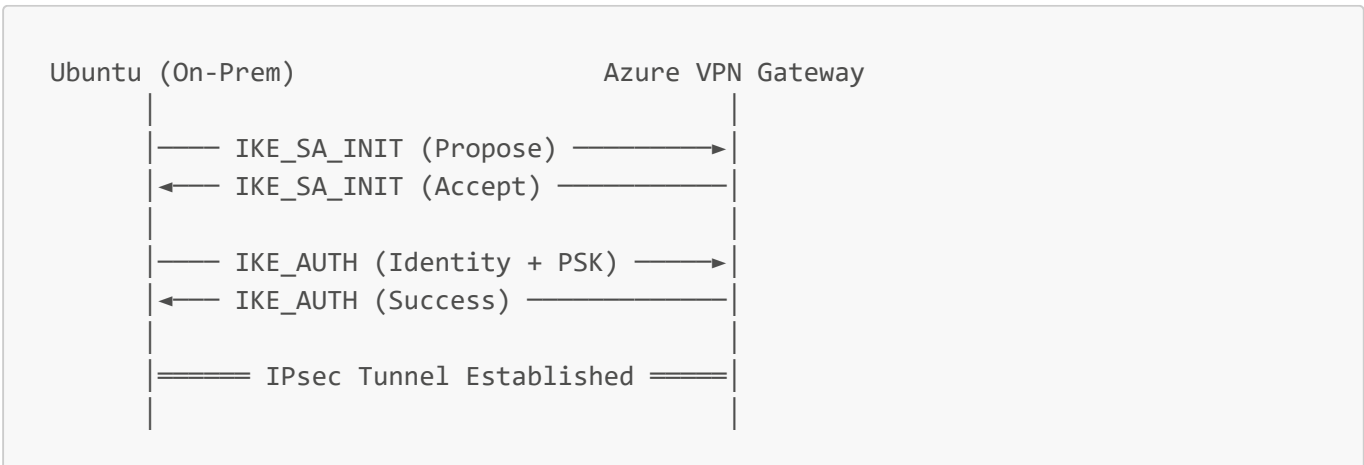
# Step 16: Get public IP (needed for Azure configuration)
curl -4 ifconfig.me
# Note this IP - you will need it for Azure Local Network Gateway
```

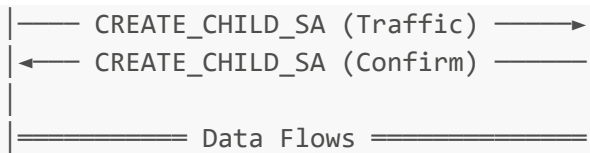
6. VPN Configuration

6.1 VPN Architecture

VPN TUNNEL DETAILS	
Protocol	: IPsec/IKEv2
Authentication	: Pre-Shared Key (PSK)
Encryption (IKE)	: AES-256 / SHA-256 / DH Group 2 (modp1024)
Encryption (ESP)	: AES-256 / SHA-256
Local Network	: 172.16.0.0/24 (On-premises)
Remote Network	: 10.0.0.0/8 (Azure VNets)
DPD	: Enabled (30s delay, 120s timeout)
Auto-reconnect	: Enabled

6.2 IKEv2 Handshake Flow





7. Azure VPN Connection

7.1 Get On-Premises Public IP

```
# Run on Ubuntu Router VM
curl -4 ifconfig.me
# Example output: 203.0.113.50
# ⚠ Note this IP - required for next step
```

7.2 Create Local Network Gateway

Azure Portal → Local Network Gateways → Create

Basic:

Name : onprem-lgw
Region : East US
Resource Group : rg-hybrid-network

Endpoint:

IP Address : <YOUR_PUBLIC_IP from curl ifconfig.me>

Address Space:

172.16.0.0/24 ← Your on-premises LAN

Tags:

Environment : Production
Purpose : OnPremises-Gateway

7.3 Create VPN Connection

Azure Portal → Virtual Network Gateways → vng
→ Connections → Add

Settings:

Name : azure-to-onprem
Connection Type : Site-to-Site (IPsec)
VPN Gateway : vng
Local Network GW : onprem-lgw
Shared Key (PSK) : <YourStrongPSKHere2024!>
IKE Protocol : IKEv2

⚠ Save the PSK - you will need it for StrongSwan

Custom IPsec Policy (Optional but Recommended):

IKE Phase 1:

Encryption : AES256
Integrity : SHA256
DH Group : DHGroup2

IKE Phase 2:

Encryption : AES256
Integrity : SHA256
PFS Group : None
SA Lifetime : 3600 seconds

7.4 Note Azure VPN Gateway Public IP

Azure Portal → Virtual Network Gateways → vng
→ Overview → Public IP Address

Note: XX.XX.XX.XX ← You need this for StrongSwan config

8. StrongSwan IPsec Setup

8.1 Install StrongSwan

```
# Step 1: Update packages
sudo apt update && sudo apt upgrade -y

# Step 2: Install StrongSwan and plugins
sudo apt install -y \
  strongswan \
  strongswan-pki \
  libcharon-extra-plugins \
  libcharon-extauth-plugins \
  libstrongswan-extra-plugins

# Step 3: Verify installation
ipsec version
# Expected: Linux StrongSwan 5.x.x
```

8.2 Configure IPsec (ipsec.conf)

```
# Backup original
sudo cp /etc/ipsec.conf /etc/ipsec.conf.backup

# Edit configuration
sudo nano /etc/ipsec.conf
```

```
# /etc/ipsec.conf
# -----
# StrongSwan IPsec Configuration
# On-Premises Ubuntu Router → Azure VPN Gateway
# -----

config setup
    charondebug="ike 2, knl 2, cfg 2, net 2, esp 2, dmn 2, mgr 2"
    uniqueids=no

# — Default Connection Parameters -----
conn %default
    ikelifetime=28800s          # IKE SA lifetime : 8 hours
    keylife=3600s              # IPsec SA lifetime : 1 hour
    rekeymargin=3m             # Rekey before expiry
    keyingtries=%forever       # Keep retrying forever
    authby=secret              # Pre-shared key auth
    keyexchange=ikev2          # IKEv2 (matches Azure)
    mobike=no                  # Disable MOBIKE

# — Azure Site-to-Site VPN Connection -----
conn azure-s2s-vpn

    # Local (On-Premises) Side
    left=%defaultroute         # Auto-detect local IP
    leftid=<YOUR_PUBLIC_IP>      # ← Replace: curl ifconfig.me
    leftsubnet=172.16.0.0/24   # On-premises LAN subnet
    leftfirewall=yes           # Enable firewall traversal

    # Remote (Azure) Side
    right=<AZURE_VNG_PUBLIC_IP> # ← Replace: Azure vng-ip
    rightid=<AZURE_VNG_PUBLIC_IP> # ← Same as right
    rightsubnet=10.0.0.0/8     # Azure VNet address space
    # Adjust if your Azure VNets are different:
    # rightsubnet=10.1.0.0/16,10.2.0.0/16,10.3.0.0/16

    # Connection Behavior
    auto=start                 # Start on boot
    closeaction=restart        # Restart if connection drops
    dpdaction=restart          # Restart on dead peer detection
    dpddelay=30s               # DPD check every 30 seconds
    dpdtimeout=120s            # DPD timeout after 120 seconds

    # Encryption Settings (Must match Azure custom policy)
```

```
ike=aes256-sha256-modp1024!    # Phase 1 : IKE encryption
esp=aes256-sha256!            # Phase 2 : ESP encryption
```

8.3 Configure Pre-Shared Key (ipsec.secrets)

```
# Edit secrets file
sudo nano /etc/ipsec.secrets
```

```
# /etc/ipsec.secrets
# -----
# IPsec Pre-Shared Key Configuration
# Format: LOCAL_ID REMOTE_ID : PSK "shared-key"
# -----

<YOUR_PUBLIC_IP> <AZURE_VNG_PUBLIC_IP> : PSK "<YourStrongPSKHere2024!>"

# -----
# ⚠ IMPORTANT:
# - PSK must EXACTLY match what you set in Azure connection
# - Replace both IP placeholders with actual values
# - Keep this file secure (chmod 600)
# -----
```

```
# Secure the file
sudo chmod 600 /etc/ipsec.secrets

# Verify permissions
ls -la /etc/ipsec.secrets
# Expected: -rw----- 1 root root
```

8.4 Configure iptables for VPN Traffic

```
# Allow IKE (VPN negotiation)
sudo iptables -A INPUT -p udp --dport 500 -j ACCEPT

# Allow NAT-Traversal (VPN behind NAT)
sudo iptables -A INPUT -p udp --dport 4500 -j ACCEPT

# Allow ESP (encrypted VPN data)
sudo iptables -A INPUT -p esp -j ACCEPT

# Allow forwarding for VPN tunnel traffic
```

```
sudo iptables -A FORWARD -i ens33 -o ens33 \  
-m policy --dir in --pol ipsec -j ACCEPT  
  
sudo iptables -A FORWARD -o ens33 \  
-m policy --dir out --pol ipsec -j ACCEPT  
  
# Save all rules  
sudo netfilter-persistent save
```

8.5 Start StrongSwan

```
# Start StrongSwan service  
sudo systemctl start strongswan-starter  
  
# Enable on boot  
sudo systemctl enable strongswan-starter  
  
# Check service status  
sudo systemctl status strongswan-starter  
  
# Bring up VPN connection  
sudo ipsec up azure-s2s-vpn  
  
# Check connection status  
sudo ipsec status  
sudo ipsec statusall
```

9. Testing & Verification

9.1 Verification Checklist

PHASE 1 - Network Basics

- ☐ ip a → ens33 has DHCP IP, ens37 has 172.16.0.1
- ☐ ip route → 3 routes visible (default, LAN, WAN subnet)
- ☐ ping 8.8.8.8 → Internet reachable
- ☐ ping google → DNS resolution working
- ☐ ifconfig.me → Public IP returned

PHASE 2 - VPN Tunnel

- ☐ ipsec status → Tunnel shows ESTABLISHED
- ☐ Azure Portal → Connection shows Connected 
- ☐ ping Azure VM IP → Reachable from Ubuntu
- ☐ ping 172.16.0.1 → Reachable from Azure VM (via Bastion)

PHASE 3 - End-to-End

- Azure VM → On-prem → Traffic flows through firewall
- On-prem → Azure VM → Traffic flows through VPN + firewall
- Bastion access → Can reach both VMs via Bastion

9.2 Verification Commands

```
# — On Ubuntu Router VM —————  
  
# Check interfaces  
ip a  
ip link show  
  
# Check routes (expect 3 entries)  
ip route show  
  
# Check VPN tunnel status  
sudo ipsec status  
sudo ipsec statusall  
  
# Test internet  
ping -c 4 8.8.8.8  
  
# Test Azure VM connectivity  
ping -c 4 <AZURE-VM-PRIVATE-IP>  
  
# Test DNS  
nslookup google.com  
  
# Monitor VPN logs live  
sudo journalctl -u strongswan-starter -f  
  
# Check iptables rules  
sudo iptables -L -n -v  
sudo iptables -t nat -L -n -v
```

```
# — On Azure VM (via Bastion) —————  
  
# Test connectivity to on-premises router  
ping -c 4 172.16.0.1  
  
# Trace route to on-premises  
traceroute 172.16.0.1  
  
# Check if traffic goes through firewall  
traceroute 172.16.0.1  
# Should pass through: Azure VM → Firewall → VPN GW → Ubuntu Router
```

9.3 Expected Test Results

Test: ping 8.8.8.8 (from Ubuntu)

PING 8.8.8.8 56 bytes of data.

64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=12.3ms 

Test: sudo ipsec status

Security Associations (1 up, 0 connecting):

azure-s2s-vpn[1]: ESTABLISHED 00:05:23 ago 

azure-s2s-vpn{1}: INSTALLED, TUNNEL 

azure-s2s-vpn{1}: 172.16.0.0/24 === 10.0.0.0/8

Test: Azure Portal Connection Status

azure-to-onprem : Connected 

Data In : XX KB

Data Out : XX KB

10. Troubleshooting Guide

10.1 Issue: Only 1 Route Showing in ip route

Symptom: ip route shows only 1 entry instead of 3

Cause 1: ens37 not configured properly in netplan

Cause 2: renderer set to NetworkManager instead of networkd

Cause 3: netplan not applied after changes

Fix:

Check netplan config

sudo cat /etc/netplan/00-installer-config.yaml

Ensure renderer is networkd (not NetworkManager)

Ensure ens37 has addresses configured

Re-apply

sudo netplan generate

sudo netplan apply

Verify

ip route show

10.2 Issue: Cannot ping 8.8.8.8

Symptom: ping 8.8.8.8 fails / times out

Cause 1: No default gateway configured

Cause 2: VMware NAT not working

Cause 3: Firewall blocking ICMP

Fix:

Check default route exists

ip route | grep default

Should show: default via 192.168.x.x dev ens33

If missing, add temporarily

sudo ip route add default via <VMWARE_NAT_GATEWAY>

Check VMware NAT service

On VMware Host: Edit → Virtual Network Editor

Ensure VMnet8 (NAT) is configured

Check iptables not blocking

sudo iptables -L INPUT -n -v

sudo iptables -L FORWARD -n -v

10.3 Issue: VPN Tunnel Not Establishing

Symptom: ipsec status shows CONNECTING but not ESTABLISHED

Cause 1: PSK mismatch between Azure and StrongSwan

Cause 2: Wrong Azure VPN Gateway IP in ipsec.conf

Cause 3: Firewall blocking UDP 500/4500

Cause 4: IKEv2 encryption mismatch

Fix:

Check logs for errors

sudo journalctl -u strongswan-starter --since "5 minutes ago"

sudo tail -f /var/log/syslog | grep charon

Verify PSK matches exactly

sudo cat /etc/ipsec.secrets

Verify Azure VPN Gateway IP

ping <AZURE_VNG_IP>

Check ports are open

sudo netstat -ulnp | grep -E '500|4500'

Restart and retry

```
sudo ipsec restart
sudo ipsec up azure-s2s-vpn
```

10.4 Issue: Can Ping Azure VPN GW but Not Azure VMs

Symptom: VPN is up but cannot reach Azure VM private IPs

Cause 1: rightsubnet in ipsec.conf doesn't match Azure VNet

Cause 2: Azure Firewall blocking traffic

Cause 3: Azure NSG blocking traffic

Cause 4: Route table missing on Azure side

Fix:

Verify rightsubnet covers your Azure VM IPs

```
sudo cat /etc/ipsec.conf | grep rightsubnet
```

Check Azure Firewall rules in portal

→ Allow on-premises (172.16.0.0/24) to Azure VMs

Check NSG rules on destination Azure VM

→ Allow inbound from 172.16.0.0/24

Check route tables in Azure

→ Ensure on-prem route points to VPN Gateway

10.5 Common Error Messages

Error: "received AUTHENTICATION_FAILED notify error"

Fix : PSK mismatch → check /etc/ipsec.secrets matches Azure PSK

Error: "establishing IKE_SA failed"

Fix : Wrong Azure VPN GW IP or UDP 500 blocked

Error: "no path found"

Fix : Check rightsubnet matches actual Azure VNet ranges

Error: "CHILD_SA closed"

Fix : Encryption mismatch → verify ike= and esp= parameters

11. Security Best Practices

11.1 Network Security

- ✓ Zero Trust: No public IPs on VMs (Bastion only)
- ✓ Defense in Depth: NSG → Firewall → VPN encryption
- ✓ Least Privilege: NSG rules allow only required ports
- ✓ Traffic Inspection: All traffic through Azure Firewall
- ✓ Encrypted Transit: IPsec tunnel with AES-256
- ✓ No Split Tunneling: All traffic routed through hub

11.2 VPN Security

- ✓ Use IKEv2 (more secure than IKEv1)
- ✓ Use AES-256 encryption
- ✓ Use strong PSK (minimum 32 characters)
- ✓ Enable DPD (Dead Peer Detection)
- ✓ Rotate PSK periodically
- ✓ Consider certificate-based auth for production

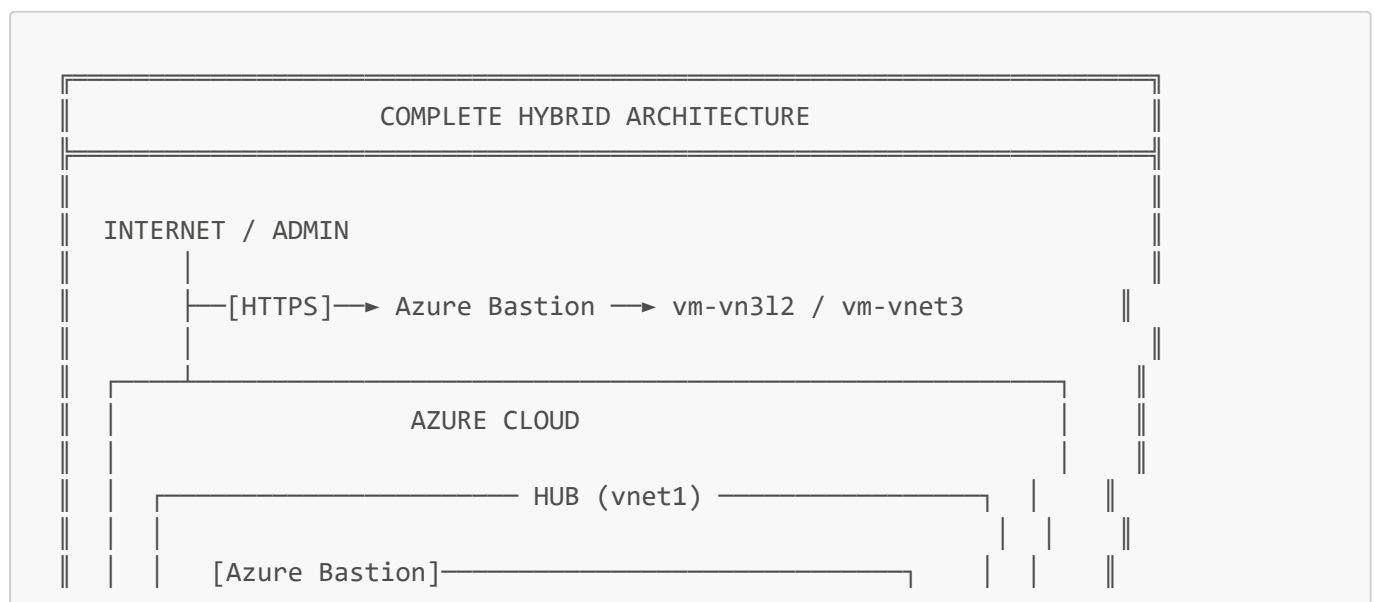
11.3 PSK Guidelines

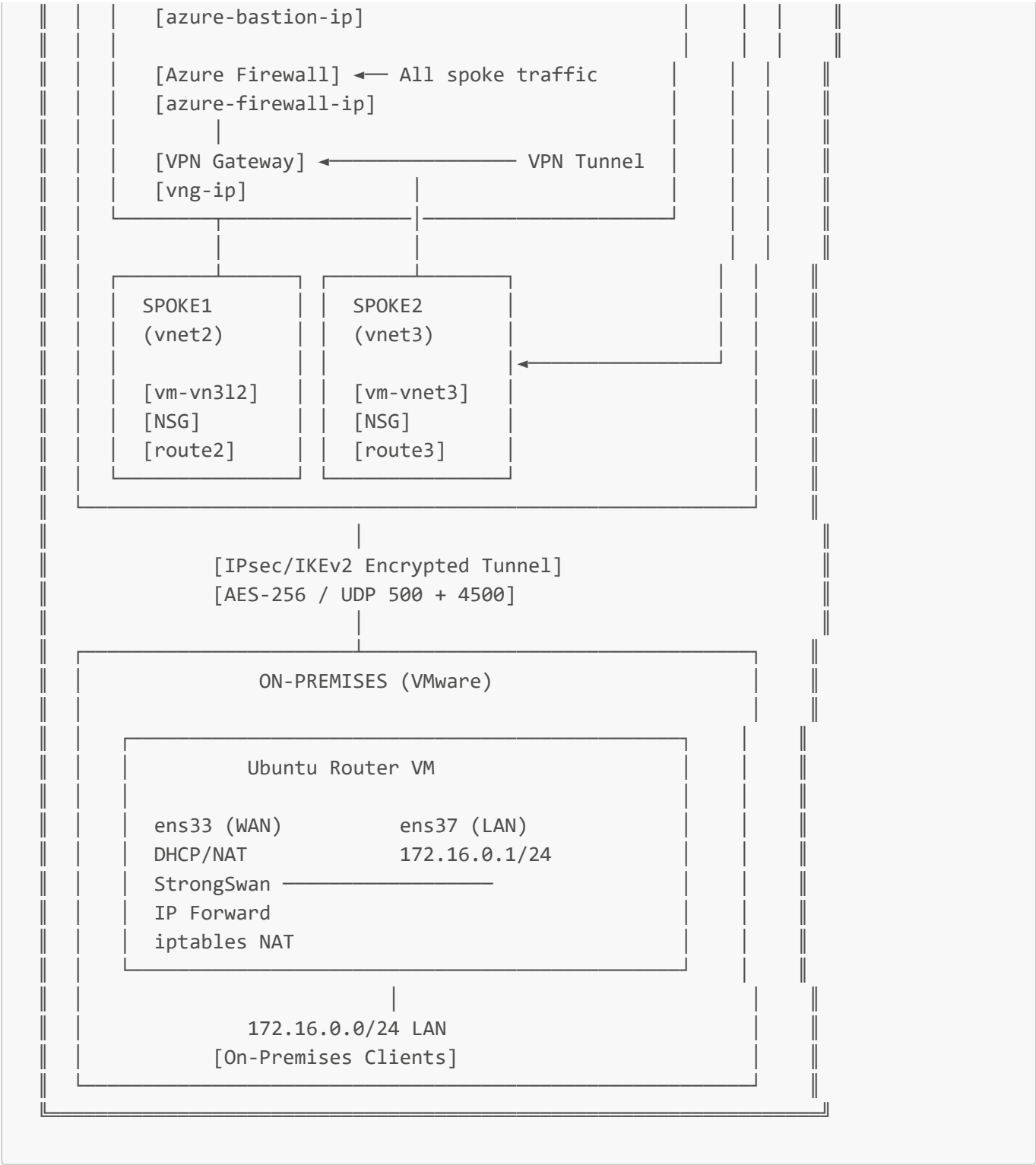
- ✗ Bad PSK : password123
- ✗ Bad PSK : azure-vpn
- ✓ Good PSK : Xk9#mP2\$vn7@qL4!wR6&tY3*
- ✓ Better : Use certificate-based authentication for production

Generate strong PSK:
`openssl rand -base64 32`

12. Architecture Summary

12.1 Final Architecture Overview





12.2 IP Address Summary

IP ADDRESS REFERENCE	
Component	IP Address / Range
vnet1 (Hub)	10.1.0.0/16
GatewaySubnet	10.1.0.0/27
AzureFirewallSubnet	10.1.1.0/26

AzureBastionSubnet	10.1.2.0/27
Azure Firewall Private	10.1.1.4
vnet2 (Spoke1)	10.2.0.0/16
workload-subnet	10.2.1.0/24
vm-vn312	10.2.1.x (DHCP)
vnet3 (Spoke2)	10.3.0.0/16
onprem-bridge-subnet	10.3.1.0/24
vm-vnet3	10.3.1.x (DHCP)
On-Premises LAN	172.16.0.0/24
Ubuntu Router (LAN)	172.16.0.1
Ubuntu Router (WAN)	DHCP (VMware NAT)
Public IP (VPN)	Dynamic (curl ifconfig.me)
Azure VPN GW Public IP	From Azure Portal (vng-ip)
Azure Bastion Public IP	From Azure Portal (azure-bastion-ip)
Azure Firewall Public IP	From Azure Portal (azure-firewall-ip)

12.3 Port & Protocol Reference

PORT & PROTOCOL REFERENCE		
Service	Protocol	Purpose
IKEv2	UDP 500	VPN key exchange
NAT-T	UDP 4500	VPN through NAT
ESP	IP Proto 50	Encrypted VPN data
SSH	TCP 22	Linux VM management (via Bastion)
RDP	TCP 3389	Windows VM management (via Bastion)
HTTPS	TCP 443	Azure Bastion portal access
ICMP	ICMP	Ping / connectivity testing
DNS	UDP 53	Name resolution
HTTP/HTTPS	TCP 80	Application traffic
	TCP 443	

12.4 Key Design Principles Summary

DESIGN PRINCIPLES

1. ZERO TRUST
 - No direct public IP on any VM
 - All access through Azure Bastion
 - Authenticated and encrypted always
2. DEFENSE IN DEPTH
 - Layer 1: NSG (per VM traffic control)
 - Layer 2: Azure Firewall (central inspection)
 - Layer 3: IPsec VPN (encrypted hybrid connectivity)
3. CENTRALIZED SECURITY
 - All inter-spoke traffic through hub firewall
 - Single point of policy enforcement
 - Centralized logging and monitoring
4. NETWORK SEGMENTATION
 - Separate VNets per workload tier
 - Controlled peering with route tables
 - On-premises isolated with VPN only
5. ENCRYPTED CONNECTIVITY
 - AES-256 IPsec tunnel
 - IKEv2 for stronger authentication
 - No unencrypted hybrid traffic



Important Notes

- **VPN Gateway Deployment** takes 30-45 minutes in Azure
- **Public IP Changes** on your ISP will break VPN - update Local Network Gateway
- **PSK Must Match Exactly** on both Azure and StrongSwan sides
- **rightsubnet** must cover all Azure VNet ranges you want to reach
- **VMware NAT** means your Ubuntu VM is behind NAT - IKEv2 NAT-T handles this
- **Production Use** should consider certificate-based VPN auth instead of PSK